Modules & The Math Module

A **module** is a file of reusable code. Built-in modules need no `pip install`.

1) What is a module?

`math` maths operations	`random` random choices	`statistics` mean, median, mode	`datetime` dates and times

2) import module: use a dot

Example :

```
import math
print("Square root of 25:", math.sqrt(25))
print("Value of pi:", math.pi)
print("Round 4.2 up:", math.ceil(4.2))
print("Round 4.8 down:", math.floor(4.8))
```

Output :

```
Square root of 25: 5.0
Value of pi: 3.141592653589793
Round 4.2 up: 5
Round 4.8 down: 4
```

the dot reaches inside the module: sqrt is a function that lives in math

3) from module import name

Example :

```
from math import sqrt, pi
print("Square root of 81:", sqrt(81))
print("Pi:", pi)
```

Output :

```
Square root of 81: 9.0
Pi: 3.141592653589793
```

Style	Write	Use it as
import module	`import math`	`math.sqrt(25)`
from ... import	`from math import sqrt`	`sqrt(25)`

Quick check

`math.ceil(4.2)` is **5** (round up) and `math.floor(4.8)` is **4** (round down).

STATISTICS & RANDOM

`statistics` summarises a list of numbers; `random` picks things by chance.

1) statistics: mean, median, mode

Example :

```
import statistics
marks = [75, 80, 92, 88, 80]
print("Marks:", marks)
print("Average:", statistics.mean(marks))
print("Median:", statistics.median(marks))
print("Mode:", statistics.mode(marks))
```

Output :

```
Marks: [75, 80, 92, 88, 80]
Average: 83
Median: 80
Mode: 80
```

mean is the average, median is the middle value, mode is the most common value

2) random: choice and randint

Example :

```
import random
random.seed(7)
students = ["Rahul", "Priya", "Amit", "Neha"]
print("Random student:", random.choice(students))
print("Random number from 1 to 6:", random.randint(1, 6))
```

Output :

```
Random student: Amit
Random number from 1 to 6: 2
```

without a seed the answers change on every run; with a seed they repeat

3) Practical task: random quiz question

Example :

```
import random
random.seed(3)
questions = ["What does def do in Python?", "What is a list?",
             "What is the purpose of return?", "What does file mode a do?"]
question = random.choice(questions)
print("Your question is:")
print(question)
```

Output :

```
Your question is:
What is a list?
```

DATETIME & COMBINING MODULES

`datetime` handles dates; modules and your own functions work together.

1) Dates and their parts

Example :

```
from datetime import date
today = date(2026, 10, 28)
print("Date:", today)
print("Year:", today.year)
print("Month:", today.month)
print("Day:", today.day)
```

Output :

```
Date: 2026-10-28
Year: 2026
Month: 10
Day: 28
```

today() gives the current date, this example fixes a date so the output repeats

2) strftime: format a date

Example :

```
from datetime import date
today = date(2026, 10, 28)
print("Report date:", today.strftime("%d-%m-%Y"))
```

Output :

```
Report date: 28-10-2026
```

Code	Meaning	Result
%d	day	28
%m	month	10
%Y	four-digit year	2026

3) A module inside your own function

Example :

```
import statistics
from datetime import date
def make_marks_report(student_name, marks):
    average = statistics.mean(marks)
    print("Student:", student_name)
    print("Marks:", marks)
    print("Average:", round(average, 2))
    print("Report date:", date(2026, 10, 28))
make_marks_report("Priya", [88, 92, 85])
```

Output :

```
Student: Priya
Marks: [88, 92, 85]
Average: 88.33
Report date: 2026-10-28
```

Practice & Revision

Write each solution yourself first, then compare. Each is run for real.

Task 1 Square root of 144 and pi.

Solution :

```
import math
print(math.sqrt(144))
print(math.pi)
```

```
12.0
3.141592653589793
```

Task 2 Mean and median of scores.

Solution :

```
import statistics
scores = [60, 70, 80, 90, 100]
print("Mean:", statistics.mean(scores))
print("Median:", statistics.median(scores))
```

```
Mean: 80
Median: 80
```

Task 3 Pick who presents first.

Solution :

```
import random
random.seed(5)
students = ["Rahul", "Priya", "Amit", "Neha"]
print("Present first:", random.choice(students))
```

```
Present first: Amit
```

Task 4 Date as DD-MM-YYYY.

Solution :

```
from datetime import date
print(date(2026, 10, 28).strftime("%d-%m-%Y"))
```

```
28-10-2026
```

Task 5 marks_summary(marks) function.

Solution :

```
import statistics
def marks_summary(marks):
    print("Average:", round(statistics.mean(marks), 2))
    print("Median:", statistics.median(marks))
marks_summary([75, 82, 91, 68, 82])
```

```
Average: 79.6
Median: 82
```

Quick reference

Module	Use	Example
`math`	maths operations	`math.sqrt(25)`
`random`	random choice or number	`random.choice(names)`
`statistics`	mean, median, mode	`statistics.mean(marks)`
`datetime`	dates and times	`date.today()`